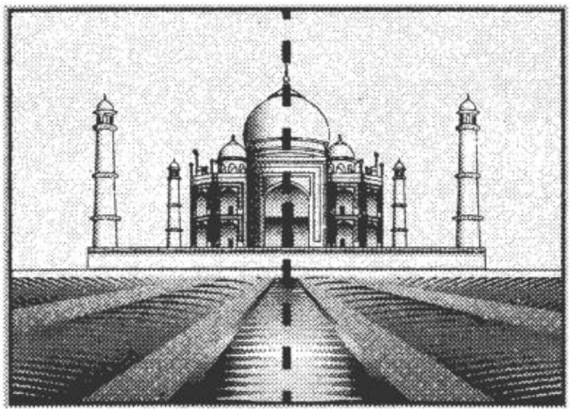
Geometry

Real – Life Examples

- ❖ The intersection of four roads on a traffic single is an example of right angle.
- ❖ The Taj Mahal is a fine example of symmetry.



LEARNING OBJECTIVES

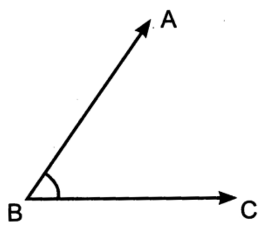
This lesson will help you to:—

- ❖ Be able to identify and classify angles as acute, right and obtuse angles.
- ❖ Be able to identify right angles in real world situations.
- ❖ Explore perspective while drawing 3D object in 2D.
- ❖ Be able to. explore rotations and reflection in 2D shapes.
- ❖ Be able to explore symmetry and nets in 3D shapes.

QUICK CONCEPT REVIEW

An angle is a measure of a turn. It is measured in degrees (°). There are different types of angles:

Acute angle: If the measure of an angle is less than 90^0, then it is called an acute angle.

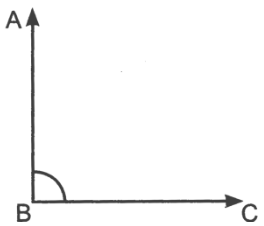


$\angle ABC$ is an acute angle.

Right angle: If the measure of an angle is 90°, then it is called a right angle.

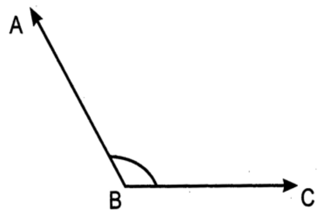
Amazing Facts

- ARCHITECTURE is based on angles and lines. They use the then bring the building to life.
- Things that are shaped like cube are often referred as 'cubic'
- most dice are in the shape of a cube with numbers 1 to 6 on each face.
- 20^0 is approximately the width of a handspan at arm's length



$\angle ABC$ is a right angle.

Obtuse angle: If the measure of an angle is greater than 90^0, then it is called an obtuse angle.



$\angle ABC$ is an obtuse angle.

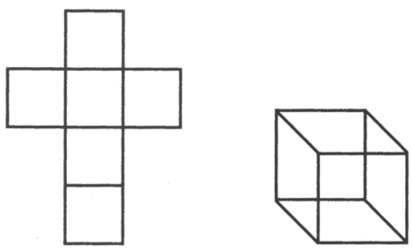
Symmetry is when one shape becomes exactly like another if it is rotated, reflected or translated.

The vertical line in the figure above is called line of symmetry.

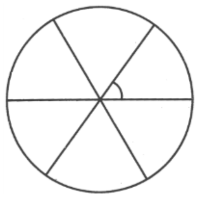
3D shapes have faces, edges and vertices. A net of a

3D shape is a figure which folds up to form a 3D shape.

For example:



Net of a cube folds up to form a cube.



Historical preview

- ❖ The first know instrument for measuring angles was the Egyptian Groma. It consisted of 4 stones hanging by cords from sticks set at right angles.

Misconception/concept

Misconcept: Students tend to find more lines of symmetry than actually exist.

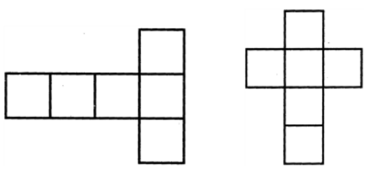
Concept: The line of symmetry divides the object such that it forms two parts that are mirror image of each other.

Concept: An angle of measure 90^0 is called right angle. It has nothing to do with direction.

Properties:

- ❖ There are 360^0 in a full turn.
- ❖ A protractor can be used to measure an angle.
- ❖ Angles can be observed all around us.
- ❖ A shape can be turned clockwise or counter clockwise about a point. It is then said to be rotated.
- ❖ A shape can be reflected in a mirror line.
- ❖ There are several possible nets of a 3D shape.

For example:



These are two different nets that form cubes.

GAME

1. Get two match sticks from a match box. The teacher creates angles of different measures from the sticks. The teacher then asks the students to classify the angles as acute, right or obtuse angles.
2. The teacher will show different objects in class such as pencil box, eraser, book etc., and ask the students if the objects have a line of symmetry.